

CHAPTER ONE

Audit Sampling

Chapter Objectives

At the end of this chapter, students will be able to:

- ✓ Define audit sampling
- ✓ Identify the type of audit procedures that do not involve audit sampling
- ✓ Distinguish between sampling risk and non-sampling risk
- ✓ Discuss the different types of audit sampling design
- ✓ Apply statistical and non-statistical sampling methods to tests of controls and substantive tests.

Nature of Audit Sampling

- ✓ Sampling refers to the process of selecting a group of items from a large group of items. The assumption of sampling is that the sample is the representative of the population.
- ✓ Audit sampling refers to the application of an audit procedure to less than 100% of the items within an account balance and class of transactions for the purpose of evaluating some characteristic of the balance or class.
- ✓ Auditors may encounter two types of risk in evidence gathering process:
 1. **Sampling risk:**
 - the possibility that the sample drawn is not representative of the population and, as a result, the auditor will reach an incorrect conclusion.
 - Is a function of sample size. The larger the sample size, the lower the sampling risk and vice versa.
 - can be quantified using statistical sampling.
 - can be reduced by increasing sample size
 - Key issue: balance sampling risk with the cost of using large sample
 2. **Non-sampling risk:** the possibility that the auditor uses inappropriate audit procedures, fail to detect a misstatement when applying an audit procedure, or misinterpret an audit result. It can not be quantified using statistical sampling. The uncertainty related to such risk can be controlled by:
 - Adequate training
 - Proper planning
 - Effective supervision
- ✓ Audit sampling is needed for audit test. However, all audit tests do not involve audit sampling. Which audit tests do not involve sampling?

Audit Tests that do not involve sampling

- a. *Inquiry and Observation*
 - Are used extensively as a source of evidential matter such as:
 - ✓ To understand the components of internal control
 - ✓ To evaluate many of the inherent risk factors
 - ✓ To establish the existence of some items
- b. *Analytical procedures*
 - Especially simple analytical procedures such as simple comparisons and ratio analysis
- c. Procedures applied to every item in the population
- d. *Classes of transactions or account balances not tested*
 - Because of an acceptably low risk of material misstatement or immateriality
- e. Tests of automated information technology controls

Type I and Type II Errors

A. In Relation to Tests of controls

1. Type I Error (alpha)
 - The risk that the assessed level of control risk based on the sample is greater than the true operating effectiveness of the control
 - Risk of assessing control risk too high
2. Type II Error (beta)
 - The risk that the assessed level of control risk based on the sample is less than the true operating effectiveness of the control
 - Risk of assessing control risk too low

B. In relation to Substantive tests

1. Type I Error (alpha)
 - The risk that the sample supports the conclusion that the recorded account balance is materially misstated when it is not materially misstated
 - Risk of incorrect rejection
 - Relates to the **efficiency** of the audit (more audit work than necessary)
2. Type II Error (beta)
 - The risk that the sample supports the conclusion that the recorded account balance is not materially misstated when it is materially misstated
 - Risk of incorrect acceptance
 - Relates to the **effectiveness** of the audit (less audit work than necessary and may lead to litigation against the auditor)

Types of Audit Sampling

1. Non-statistical (or Judgmental) sampling

- ✓ The auditor considers sampling risk when evaluating the results of an audit sample without using statistical theory to measure sampling risk
- ✓ Requires the auditor's professional judgment to plan, perform, and evaluate the sample evidence.
- ✓ Limitation: may not be as effective as statistical sampling.

2. Statistical Sampling

- Uses the laws of probability to select and evaluate the result of audit sample
- Helps the auditor:
 - ✓ Design an efficient sample
 - ✓ Measure the sufficiency of evidence obtained
 - ✓ Quantify sampling risk
- Limitations (drawbacks): Involve additional costs of:
 - ✓ Training auditors in the proper use of sampling techniques
 - ✓ Designing and conducting the sampling application
- Three major types of statistical sampling plans may be used:
 - a. Attribute sampling
 - ✓ Used to estimate the proportion of population that possesses specific characteristic.
 - ✓ Is commonly used for tests of controls
 - b. Monetary-Unit Sampling (MUS) or Probability-Proportional-to-Size Sampling (PPS)
 - ✓ Uses attribute sampling theory and techniques to estimate the birr amount of misstatement for a class of transactions or an account balance.
 - ✓ designed to test for overstatements
 - ✓ Its sampling unit is individual Dollar (Birr)
 - ✓ is important for auditing accounts receivable, loans receivables, investment securities, and inventory
 - ✓ estimates the percentage of monetary units in a population that might be misstated and then multiplies this percentage by an estimate of how much the dollars are misstated.
 - ✓ Advantages of MUS:
 - 1. When the auditor expects no misstatement, MUS usually results in a smaller sample size than classical variables sampling.
 - 2. The calculation of the sample size and evaluation of the sample results are not based on the variation between items in the population.
 - 3. When applied using the probability-proportional-to-size procedure, MUS automatically results in a stratified sample.
 - ✓ Disadvantages of MUS:
 - 1. The selection of zero or negative balances generally requires special design consideration.
 - 2. The general approach to MUS assumes that the audited amount of the sample item is not in error by more than 100%.
 - 3. When more than one or two misstatements are detected, the sample results calculations may overstate the allowance for sampling risk.
 - c. Classical Variables Sampling (CVS)
 - ✓ Used to determine if a class of transactions or account balance is materially misstated and aims at the rate of misstatement.
 - ✓ uses normal distribution theory to evaluate the characteristics of a population based on sample data. Auditors most commonly use classical variables sampling to estimate the size of misstatement
 - ✓ Its sampling unit is each transaction or account balance
 - ✓ Advantages of CVS:

1. When the auditor expects a large number of differences between book and audited values, this method will result in smaller sample size than MUS.
 2. The techniques are effective for both overstatements and understatements.
 3. The selection of zero balances generally does not require special sample design considerations.
- ✓ Disadvantages of CVS:
1. Does not work well when little or not misstatement is expected in the population.
 2. To determine sample size, the auditor must estimate the standard deviation of the audited value or differences.
 3. If few misstatements are detected in the sample data, the true variance tends to be underestimated, and the resulting projection of the misstatements to the population is likely not to be reliable.

Sample Selection Methods

- ✓ The sample should be the representative of the population
- ✓ All items must have the opportunity to be selected
- ✓ May use the following sample selection methods:
 - a. **Random selection** (the use of random number table and can be used for both non-statistical and statistical sampling)
 - b. **Systematic selection** (requires the determination of sampling interval and a starting number, and can be used for both non-statistical and statistical sampling)
 - c. **Haphazard selection** (selection of sampling units without any conscious bias and used for non-statistical sampling)
 - d. **Stratified sampling** (dividing the population in to strata and select a sample from each stratum)
 - e. **Block sampling** (the sample consists of all items in a selected time period, numerical sequence, or alphabetical sequence)

Performance and Evaluation

- Must address four issues:
 - a. The auditor must consider the effect of not being able to apply a planned audit procedure to a sampled item
 - b. The auditor should project the sample results to the population being tested and compare those results with the planned amounts
 - c. The auditor must give appropriate consideration to sampling risk
 - d. The auditor must adequately consider qualitative aspects of misstatements such as:
 - ✓ The nature and causes of misstatements
 - ✓ The possible relationship of the misstatement to other phases of the audit

Application of Audit (attribute) Sampling to Tests of Control

Attribute sample is concerned with one-sided test because the auditor is generally concerned with the maximum deviation rate in the population.

Step 1: State the objectives of the audit test.

- ✓ The objective is to evaluate the operating effectiveness of internal control

Step 2: Define the control deviation conditions

- ✓ A deviation is a departure from adequate performance of the internal control
- ✓ It is necessary to clearly define the situation in which deviation occurs.
Eg. Let's say the objective of the test is to check whether purchase invoice was approved by the finance manager before being recorded. A purchase order which was not approved (no signature & stamp) by the finance manager is considered deviation.

Step 3: Define the population & sampling units

- ✓ **Population:** Population depends on the objective of the test. If the objective of the test is to check whether purchase invoice was approved by the finance manager before being recorded, all purchase orders received during the period are considered population. The auditor should ensure that the sample frame is complete.
- ✓ **Sampling unit:** Sampling unit refers to the individual members of the population. A sampling unit may be a document, an entry or a line item. In the above example, the sampling unit is individual purchase invoice.

Step 4: Determine the sample size

- ✓ Factors to be considered in determining the sample size:

1. Determining the acceptable risk of assessing control risk too low

- the risk that the sample will support the auditor's planned degree of reliance on the control when the true deviation rate for the population does not justify such reliance (type II error)
- Its determination is primarily based on the auditor's professional judgment
- the auditor should consider the importance of the audit objective
- There is an inverse relationship between the risk of assessing the risk control risk too low and sample size
- The smaller the risk of assessing control risk too low, the larger the sample size must be.
- Factors to be considered in setting the acceptable risk of assessing control risk too low:
 - a. Effectiveness concern
 - b. Efficiency concern

2. The tolerable deviation rate

- the maximum deviation from a prescribed control that the auditor is willing to accept without altering the planned assessed level of control risk
- The suggested tolerable deviation rates for assessed levels of control risk

Planned assessed level of control risk	Tolerable deviation rate
Low	3 - 7%
Moderate	6 - 12%
Slightly below maximum	11 - 20%
Maximum	Omit test

- The tolerable deviation rate is inversely related to the sample size
- The lower the tolerable deviation rate, the larger the sample size

- The concern of the auditor is whether the true deviation rate exceeds the tolerable deviation rate
- There is direct relationship between the planned assessed level of control risk and tolerable deviation rate

3. The expected population deviation rate

- is the deviation rate that the auditor expects to exist in the population
- can be developed based on prior years' results, past experience from similar tests on other engagements, or on a pilot sample
- The expected population deviation rate has a direct relationship to sample size.
- The larger the expected population deviation rate, the larger the sample size must be.
- If the auditor believes that the expected population deviation rate exceeds the tolerable deviation rate, the statistical test should be omitted and substantive tests should be used.

The effect of Sample Selection Factors on Sample Size			
Factors	Relationship to sample	Examples	
		Change in Factor	Effect on Sample
Acceptable risk of assessing control risk too low	Inverse	Lower Higher	Increase Decrease
Tolerable Deviation rate	Inverse	Lower Higher	Increase Decrease
Expected Population deviation rate	Direct	Lower Higher	Decrease Increase

Once the above three factors are determined, the appropriate sample size is determined from the table shown on the following page:

TABLE 8-5

Statistical Sample Sizes for Attribute Sampling—95 Percent Desired Confidence Level (i.e., 5 Percent Risk of Incorrect Acceptance)

Expected Population Deviation Rate	Tolerable Deviation Rate										
	2%	3%	4%	5%	6%	7%	8%	9%	10%	15%	20%
0.00%	149(0)	99(0)	74(0)	59(0)	49(0)	42(0)	36(0)	32(0)	29(0)	19(0)	14(0)
.25	236(1)	157(1)	117(1)	93(1)	78(1)	66(1)	58(1)	51(1)	46(1)	30(1)	22(1)
.50	*	157(1)	117(1)	93(1)	78(1)	66(1)	58(1)	51(1)	46(1)	30(1)	22(1)
.75	*	208(2)	117(1)	93(1)	78(1)	66(1)	58(1)	51(1)	46(1)	30(1)	22(1)
1.00	*	*	156(2)	93(1)	78(1)	66(1)	58(1)	51(1)	46(1)	30(1)	22(1)
1.25	*	*	156(2)	124(2)	78(1)	66(1)	58(1)	51(1)	46(1)	30(1)	22(1)
1.50	*	*	192(3)	124(2)	103(2)	66(1)	58(1)	51(1)	46(1)	30(1)	22(1)
1.75	*	*	227(4)	153(3)	103(2)	88(2)	77(2)	51(1)	46(1)	30(1)	22(1)
2.00	*	*	*	181(4)	127(3)	88(2)	77(2)	68(2)	46(1)	30(1)	22(1)
2.25	*	*	*	208(5)	127(3)	88(2)	77(2)	68(2)	61(2)	30(1)	22(1)
2.50	*	*	*	*	150(4)	109(3)	77(2)	68(2)	61(2)	30(1)	22(1)
2.75	*	*	*	*	173(5)	109(3)	95(3)	68(2)	61(2)	30(1)	22(1)
3.00	*	*	*	*	195(6)	129(4)	95(3)	84(3)	61(2)	30(1)	22(1)
3.25	*	*	*	*	*	148(5)	112(4)	84(3)	61(2)	30(1)	22(1)
3.50	*	*	*	*	*	167(6)	112(4)	84(3)	76(3)	40(2)	22(1)
3.75	*	*	*	*	*	185(7)	129(5)	100(4)	76(3)	40(2)	22(1)
4.00	*	*	*	*	*	*	146(6)	100(4)	89(4)	40(2)	22(1)
5.00	*	*	*	*	*	*	*	158(8)	116(6)	40(2)	30(2)
6.00	*	*	*	*	*	*	*	*	179(11)	50(3)	30(2)
7.00	*	*	*	*	*	*	*	*	*	68(5)	37(3)

*Sample size is too large to be cost-effective for most audit applications. The number in parentheses represents the maximum number of deviations in a sample of that size that allows the auditor to conclude that the tolerable deviation rate is not exceeded.

Step 5: Determine sampling method and Select the sample

Once the sample size is determined, the sample will be selected using the appropriate techniques

Step 6: Perform the audit procedures/Test the sample items/

- ✓ In conducting the planned audit procedures, the following are not considered deviation
 - properly Voided documents
 - Unused or inapplicable documents
- ✓ If the auditor is unable to examine a sampled item, the sample item is considered a deviation
- ✓ If a large number of deviations are detected early in the tests of controls, the auditor should consider stopping the test because the result of the test will not support the planned assessed level of control risk

Step 7: Evaluate the Sample Result

1. Calculate the sample results

- ✓ involves summarizing the deviations by the controls tested and evaluating the results
- ✓ involves calculating the sample deviation rate and the computed upper deviation rate
 - a. **Sample deviation rate**
 - is the number of deviations in the sample divided by the number of items in the sample
 - represents the auditor's best estimate of the population's deviation rate. However the auditor must consider allowance for sampling risk
 - b. **Computed upper deviation rate**
 - is the sum of sample deviation rate and allowance for sampling risk
 - represents the upper limit for population deviation rate

2. Performing Error Analysis

- ✓ involves evaluating the qualitative aspects of the deviations identified
- ✓ Involves two considerations:
 - a. the nature of each deviation and its cause
 - Error or fraud?
 - Misunderstanding of instructions or carelessness?
 - b. the impact of the deviation on other phases of the audit

Step 8: Draw final conclusion

- ✓ involves comparing the tolerable deviation rate to the computed upper deviation rate
 - If the computed upper deviation rate is less than or equal to the tolerable deviation rate, the auditor can conclude that the controls can be relied on
 - If the computed upper deviation rate exceeds the tolerable deviation rate, the auditor may perform the following:
 1. Modify sample size
 2. Modify tolerable deviation rate
 3. Conclude that the controls are not operating at an acceptable level.
- ✓ The auditor's professional judgment of the results of tests of control affects the final conclusion.

Step 9: Document the sampling Procedures

- All steps performed so far should be documented in the auditor's working papers

Application of Audit Sampling to Substantive Tests-Monetary Unit Sampling

Step 1: State the objectives of the audit test

- When auditors sample for tests of details of balances, the objective is to determine whether the account balance being audited is fairly stated.
- Sampling may be used for substantive testing to test the reasonableness of assertions about a financial statement amount, and to develop an estimate of some amount.

Step 2: Define Misstatement Conditions

- Misstatement conditions are any conditions that represent a monetary misstatement in a sample item.
- A misstatement is defined as the difference between monetary amounts in the client's records and amounts supported by audit evidence.
- When the client's amount deviates from amount supported by audit evidence, misstatement is said to occur. However, it may not be material in amount.

Step 3: Define the population and the sampling units

- In testing for the existence objective, the recorded dollar population is the population.
- For MUS the population is defined as the monetary value of an account balance, such as accounts receivable, investment securities, or inventory
- For MUS, an individual dollar represents the sampling unit. However, the auditor tests not individual birr but rather the account or the transaction that contains the selected birr. The account balance or transaction that contains the selected Birr is called the logical unit.

Step 4: Determine the sample size

- Factors to be considered:

1. Variation within the population

- Population variation and sample size have direct relationship in classical variables sampling.
- As population variation increases, the sample size should increase
- Population variation Can be minimized by stratifying the population
- Population variation does not affect the sample size when monetary unit sampling is used because the sample selection method considers indirectly.

2. Acceptable risk of incorrect acceptance (type II error)

- There is an inverse relationship between risk of incorrect acceptance and the sample size
- The lower the risk for incorrect acceptance, the larger the sample size must be

3. Tolerable misstatement

- The tolerable misstatement is the amount of the preliminary judgment about materiality that is allocated to an account
- The tolerable misstatement is the maximum amount by which the account can be misstatement with the auditor still accepting the account as being fairly presented
- Tolerable misstatement is inversely related to sample size.
- The lower the amount of tolerable misstatement, the larger the sample size must be.

4. Expected misstatement

- The expected misstatement is the amount of misstatement that the auditor believes exists in the population.
- The expected misstatement can be developed based on the assessment of inherent risk, prior year's results, a pilot sample, the results of related substantive tests, or the results of tests of controls.
- There is a direct relationship between expected misstatement and sample size.
- The larger the expected misstatement, the larger the sample size must be

5. Population size

- Population size is directly related to sample size
- The larger the population size, the larger the sample size must be.

Determination of sample size for Monetary-Unit Sampling:

- Attribute sampling table can be used
- Steps:
 1. Determine the risk for incorrect acceptance
 2. convert the tolerable misstatement and the expected misstatement to percentages of book value
 3. Use attribute sample table

Alternatively, the sample size can be computed using the following formula:

$$n = \frac{BV \times RF}{(TM - (EM \times EF))}$$

Where,

n = sample size

BV = Book Value

TM = Tolerable Misstatement

EM = Expected Misstatement

EF = Expansion Factor

RF = Reliability Factor

*Expansion factor adjusts the expected misstatement for the risk of incorrect acceptance.

Example: Suppose the auditor has established a tolerable misstatement of Br. 250,000 and an expected misstatement of Br. 50,000 for an account receivable account with a book value of Br. 5,000,000. The risk of incorrect acceptance is 5% and the corresponding reliability factor and expansion factor (adjustment of expected misstatement for the risk of incorrect acceptance) are 3.00 and 1.6 respectively. Note that the reliability factor and expansion factor are obtained from relevant tables.

Tolerable misstatement rate = $250,000 / 5,000,000 = 0.05$

Expected misstatement = $50,000 / 5,000,000 = 0.01$

Sample size (n)

$$\text{Sample size (n)} = \frac{5,000,000 \times 3}{(250,000 - (50,000 \times 1.6))}$$

n = 88

Step 5: select the sample items using systematic sampling method

For Monetary-Unit Sampling, (by applying systematic sampling method) probability-proportional-to-size method is used, which gives each birr in the population an equal chance of being selected and divides the population in to equal groups of birr. Each group of birr represents a sampling interval, which is obtained by dividing the book value of the population by the sample size

Assume a client's book value of accounts receivable is Br.5,000,000, and the auditor determined a sample size of 88. The sampling interval will be Br. 56,818. The auditor randomly selects a number between 0 and 56,818 and then select every 56,818th Birr to identify the balance to be included in the sample.

Step 6: Perform the audit procedures

- After the sample items have been selected, the auditor conducts the planned audit procedures on the logical units containing the selected dollar sampling units.

Step 7: Evaluate the sample results

1. Calculate the sample results

- projecting the misstatement detected in the sample to the population (called misstatement limit), which represents the mean misstatement in the population
- calculating an allowance for sampling risk and add it to the projected misstatement in order to determine the misstatement limit

2. Perform error analysis

Step 8: Draw final conclusions

- Tolerable misstatement is compared to the sum of projected misstatement and allowance for sampling risk (misstatement limit)
- If the sum of projected misstatement and allowance for sampling risk is less than or equal to tolerable misstatement, the auditor can conclude that the account is not materially misstated for the tested assertion
- The sum of projected misstatement and allowance for sampling risk is greater than tolerable misstatement, the auditor can conclude that the account is materially misstated for the tested assertion. When faced with this situation, the auditor may:
 1. Increase the sample size.
 2. Perform other substantive procedures.
 3. Request the client adjust the accounts balance.
 4. If the client refuses to adjust the account balance, the auditor would consider issuing a qualified or adverse opinion.

Step 9: Document the sampling procedures

Illustration on Monetary Unit Sampling Method

The auditors wish to test the valuation of accounts receivable in the audit of ABC Company with the objective of ascertaining whether the balance is fairly stated. The client has Br. 2,500,000 of total recorded receivables, composed of 500 accounts. The auditors have determined the following:

Tolerable misstatement (error)	Br. 50,000
Risk of incorrect acceptance	10% (Reliability Factor = 2.31)
Expected misstatement (error)	Br. 20,000 (Expansion factor = 1.50)

The auditors have decided to use Probability-proportional-to-Size Sampling. Assume that the auditors have tested the sample and discovered the following four misstatements:

Book value (BV)	Audited Value (AV)
Br. 1000	920
4000	3500
5000	4800
9800	9000

Reliability factors for errors of overstatement (partial)

Number of overstatement errors	Risk of incorrect Acceptance		
	1%	5%	10%
0	4.61	3.0	2.31
1	6.64	4.75	3.89
2	8.41	6.30	5.33
3	10.05	7.76	6.69
4	11.61	9.61	8.00
5	13.11	10.52	9.28

Expansion Factors (EF) for expected error

	Risk of Incorrect Acceptance								
	1%	5%	10%	15%	20%	25%	30%	37%	50%
EF	1.9	1.6	1.5	1.4	1.3	1.25	1.2	1.15	1.0

Required:

- Compute the following:
 - the required sample size
 - sampling interval
 - projected misstatement
 - Basic precision
 - Incremental allowance
 - Upper limit on misstatement
- Draw final conclusion about the fairness of accounts receivable balance

Solution**1. Computations**

- Sample size = $\frac{2,500,000 \times 2.31}{50,000 - (20,000 \times 1.5)} = \underline{289}$
- Sampling interval = $2,500,000 / 289 = \underline{8651}$
- Projected misstatement:
 - ✓ Projected misstatement in the population is the sum of projected misstatement for each account identified as misstatement. It can be computed by considering the following cases:
 Case 1: When BV of the misstated account is less than the sampling interval
 Projected misstatement = Tainting Percentage x Sampling interval
 Tainting percentage = the percent of misstatement in the account
 = $(BV - AV) / BV$
 Case II: When BV of the misstated account is equal to or greater than the sampling interval
 Projected misstatement = the actual misstatement in the account
 = $BV - AV$

Thus, total projected misstatement in the population = case 1 + case 2

BV	Audited value	Misstatement	Tainting %	Sampling interval	Projected misstatement
Br. 1000	920	80	0.08	8651	692
4000	3500	500	0.125	8651	1081
5000	4800	200	0.04	8651	346
9800	9000	800	NA	NA	800
19,800	18,220	1580			2919

Most Likely Misstatement (MLM) - The best estimate of the actual amount of dollar misstatements that exist in the account balance. This is also called projected misstatement.

- Basic precision = Reliability factor X sampling interval
 = $2.31 \times 8651 = 19,984$
 *Basic precision is the minimum allowance for sampling risk, given no errors or misstatements
- Incremental allowance
Incremental Allowance for Sampling Error - An increase in the upper misstatement estimate caused by the statistical properties of misstatements found.

Steps:

- ✓ Rank the projected misstatement for accounts with BV less than the sampling interval (from largest to smallest)
- ✓ Multiply each projected misstatement by the incremental reliability factor, reduced by one
- ✓ Sum the resulting amounts to get Incremental Allowance

Reliability factor	Increment	Increment - 1	Projected	Incremental
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			misstatement	Allowance
2.31	-	-	-	-
3.89	1.58	0.58	1081	627
5.33	1.44	0.44	692	305
6.69	1.36	0.36	346	125
				1057

g. Upper limit on misstatement = projected misstatement + Basic precision + Incremental allowance

$$= 2919 + 19,984 + 1057 = 23,960$$

2. Draw final conclusion

Decision rule: If the upper limit on misstatement is less than or equal to the tolerable misstatement, we conclude that the balance is not materially misstated. Since upper limit on misstatement of Br.23,960 is less than tolerable misstatement of Br. 50,000, we can conclude that account receivable balance is not materially misstated.

If the upper limit on misstatement were greater than the tolerable misstatement, the auditor may:

- ✓ Increase the sample size.
- ✓ Perform other substantive procedures.
- ✓ Request the client adjust the accounts receivable balance.
- ✓ If the client refuses to adjust the account balance, the auditor would consider issuing a qualified or adverse opinion.

Note

- The expansion factor has direct effect on sample size
- The smaller the specified risk of incorrect acceptance, the larger the expansion factor
- When the sample includes no misstatements, the projected misstatement is equal to zero. Thus, upper limit on misstatement consists only of **basic precision**.
- For calculating the sample size, the zero misstatement row of the reliability factor table is always used.
- Allowance for sampling risk is not provided for book value in excess of sampling interval because the entire balance is audited
- When understatement errors are detected, the auditor can calculate a NET upper limit on misstatement. Procedures:
 - ✓ Compute projected misstatement for understatements
 - ✓ Subtract project misstatement of understatements from upper limit on overstatements
 - ✓ The result is Net upper limit on misstatement
 - ✓ Use Net upper limit on misstatement for conclusions

EXERCISES

Exercise 1

Abebe, ACCA, was conducting the audit of ABC Company for the year ended December 31, 2010. He plans to use Monetary unit sampling to audit the client's inventory balance. The balance at December 31, 2010 was Br 9,000,000. Consider the following data:

- Tolerable misstatement = 360,000 Birr
- Expected misstatement = 90,000
- Risk of incorrect acceptance = 5%

Abebe has identified the following misstatements after conducting sampling testing:

Misstatement Number	Book value	Audited value
1	10,000	7500
2	9,000	6000
3	60,000	0
4	800	640

Required:

1. compute the following:
 - a. the sample size.
 - b. the sampling interval.
 - c. projected misstatement (error)
 - d. basic precision
 - e. incremental allowance
 - f. Upper limit on misstatement
2. What conclusion should Abebe make about inventory account?
3. Assume that, in addition to the four misstatements identified above, Abebe has identified the following two understatement:

Misstatement Number	Book value	Audited value
5	6,000	6500
6	750	800

Compute the net upper limit on misstatement

Exercise 2

Suppose you are assigned to audit the accounts receivable balance of Alem Company. You decide to use statistical sampling to test the reasonableness of the recorded book value. Because the company's internal control procedures over accounts receivable have been evaluated by you as excellent and you believe few errors will be found, you decide to use probability proportional-to-size sampling. The company has 5,000 customer accounts with a total book value of Br.1, 700,000. You decide Br.75, 000 is the maximum tolerable misstatement and anticipate that there may be Br.15, 000 of error in the population.

You wish to limit the risk of incorrect acceptance to 5%. It is your intention to seek positive confirmation of accounts included in your sample and to apply alternative procedures to accounts for which no reply is received. Based on your procedures, the following errors were found in the sample:

Sample item	1	2	3	4	5
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Book value	400	750	20,000	1000	7000
Audited Value	300	705	16,000	1200	7500

Reliability factors for errors of overstatement (partial)

Number of overstatement errors	Risk of incorrect Acceptance		
	1%	5%	10%
0	4.61	3.0	2.31
1	6.64	4.75	3.89
2	8.41	6.30	5.33
3	10.05	7.76	6.69
4	11.61	9.61	8.00
5	13.11	10.52	9.28

Expansion Factors (EF) for expected error

	Risk of Incorrect Acceptance								
	1%	5%	10%	15%	20%	25%	30%	37%	50%
EF	1.9	1.6	1.5	1.4	1.3	1.25	1.2	1.15	1.0

Required:

- Compute the sample size.
- Compute the sampling interval.
- Compute projected misstatement (error)
- Compute basic precision
- Compute incremental allowance
- Upper limit on misstatement
- State your conclusion based on the above analysis

Exercise 3

The external auditor of Sheger Corporation has been testing the effectiveness of the design and implementation of Lagar Corporation's (the client) internal control system concerning all purchase invoices (100 invoices) received from suppliers during the year ended December 31, 2010. From past experience and his professional judgment, he set the risk of assessing control risk too low (type II error) and tolerable deviation rate to be 5% and 12% respectively. He selected a sample of 15 purchase invoices and tested for the agreement between purchase invoice and the related purchase order and receiving reports, and found that 3 purchase invoices were not supported by purchase orders (deviations from the prescribed control). He does not believe that the sample of 15 purchase invoices completely represent the population of purchase invoices. As a result, he provided for allowance for sampling risk of 2%.

Required:

- Determine sample deviation rate
- Compute upper deviation rate
- Should the internal control system over purchases be relied on? If not, why?